RESOLVING AT A HEIGHT-ZERO PRIME VIA A NILPOTENT-SPREAD FACTORIZATION

THE STACKS PROJECT AUTHORS

Editorial note. Input lemma-delocalize-weak is a published entry (tag 07F9) used as a standing input by its display number; not proven on the author's behalf.

Editorial note. The source uses the final-solve and standardization lemmas tacitly through the delocalize-weak conclusion; they are carded at L26797, L26799, L26801 and referenced there.

Allowed standard prerequisites (editorial).

- Weak delocalization of a localized resolution. lemma-delocalize-weak (same chapter, PUBLISHED 07F9). Produces a finitely presented factorization A -> C -> Lambda with singular ideal not inside q.
- Solving when the singular ideal generates the unit ideal. lemma-final-solve (carded L26797). Used in the background: the localized resolution may be taken smooth.
- Standardization of a smooth algebra. lemma-smooth-standard-smooth (carded L26799). Standardizes the smooth C in the delocalized factorization.
- Standard smooth presentation with prescribed coordinates. lemma-standard-smooth-include-generators (carded L26801). Chooses the presentation of C containing the coordinates of A.

Source reference. Original lemma 07F9: sources/stacks-smoothing/pinned-original-source.tex, source lines 2365–2365. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07F7: sources/stacks-smoothing/pinned-original-source.tex, source lines 2296–2296. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $R \to A \to \Lambda \supset \mathfrak{q}$ be as in Situation 07F7. Let $\mathfrak{p} = R \cap \mathfrak{q}$. Assume*

(1) *$\mathfrak{q}$ is minimal over $\mathfrak{h}_A$,*
(2) *$R_\mathfrak{p} \to A_\mathfrak{p} \to \Lambda_\mathfrak{q} \supset \mathfrak{q}\Lambda_\mathfrak{q}$ can be resolved, and*
(3) *$\dim(\Lambda_\mathfrak{q}) = 0$.*

Then $R \to A \to \Lambda \supset \mathfrak{q}$ can be resolved.

Proof. By (3) the ring $\Lambda_\mathfrak{q}$ is Artinian local hence $\mathfrak{q}\Lambda_\mathfrak{q}$ is nilpotent. Thus $(\mathfrak{h}_A)^N\Lambda_\mathfrak{q} = 0$ for some $N > 0$. Thus there exists a $\lambda \in \Lambda$, $\lambda \notin \mathfrak{q}$ such that $\lambda(\mathfrak{h}_A)^N = 0$ in Λ.

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Say $H_{A/R} = (a_1, \ldots, a_r)$ so that $\lambda a_i^N = 0$ in Λ. By Lemma 07F9 we can find a factorization $A \to C \to \Lambda$ with C of finite presentation such that $\mathfrak{h}_C \not\subset \mathfrak{q}$. Write $C = A[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$. Set

$$B = A[x_1, \ldots, x_n, y_1, \ldots, y_r, z, t_{ij}]/(f_j - \sum y_i t_{ij}, z y_i)$$

where t_{ij} is a set of rm variables. Note that there is a map $B \to C[y_i, z]/(y_i z)$ given by setting t_{ij} equal to zero. The map $B \to \Lambda$ is the composition $B \to C[y_i, z]/(y_i z) \to \Lambda$ where $C[y_i, z]/(y_i z) \to \Lambda$ is the given map $C \to \Lambda$, maps z to λ, and maps y_i to the image of a_i^N in Λ.

We claim that B is a solution for $R \to A \to \Lambda \supset \mathfrak{q}$. First note that B_z is isomorphic to $C[y_1, \ldots, y_r, z, z^{-1}]$ and hence is smooth. On the other hand, $B_{y_\ell} \cong A[x_i, y_i, y_\ell^{-1}, t_{ij}, i \neq \ell]$ which is smooth over A. Thus we see that z and $a_\ell y_\ell$ (compositions of smooth maps are smooth) are all elements of $H_{B/R}$. This proves the lemma. $\square$